

Complementary ancestries: historical immigrants' descendants and local wages in Brazil*

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Abstract

This paper examines how the inter-regional spread of the descendants of the immigrants who arrived in Brazil between 1850 and 1960 relates to current labor market outcomes. We use linked employer–employee data, apply a surname-based classification to identify workers with non-Iberian ancestry (i.e., *descendants*), and estimate wage regressions with individual and municipality fixed effects, complemented by an instrumental variable. Descendant concentrations differ sharply between the states that hosted the historical immigrant settlements and the remaining states, where non-Iberian immigration was negligible and where labor markets were shaped by internal migration after 1960. In the latter group of municipalities, a higher local concentration of descendants is associated with higher wages; in the former, the association is much weaker. In both cases the higher wage is confined to the predominant population group with Iberian surnames, whereas workers of non-Iberian ancestry show no such wage difference. These patterns are consistent with a simple framework in which descendants and non-descendants are imperfect substitutes in production and generate positive productivity spillovers. We conclude that immigration can have long-lasting, positive impacts on the local population.

KEYWORDS: historical immigration, internal migration, descendants, surnames

JEL CODES: J15, J61, O15, R23, N36

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The analysis draws on confidential administrative data from the Brazilian Ministry of Labor and Employment, which cannot be shared. Interested researchers may request access through the Ministry. Upon acceptance, we can provide all other dataset components assembled from public sources, as well as the complete replication code.

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1 Introduction

Historical events have influence on present economic development (Nunn, 2009; Spolaore and Wacziarg, 2013). A common theme in this literature is the long-lasting impacts of migration from Eurasia to different parts of the world, from the beginning of the 16th century in various colonizing incursions (Acemoglu et al., 2001) to the so-called Age of Mass Migration in the late 19th and early 20th centuries (Hatton and Williamson, 1998). Many studies credit positive impacts to human capital, brought to the receiving countries by historical immigrants and transmitted across generations (Borjas, 1992; Rocha et al., 2017). Yet, settlers also profoundly shaped the institutions and thus the future fortune of the receiving regions, making it a challenging task to disentangle the effect of places from people (Galor et al., 2009; Acemoglu et al., 2014).

Much of this literature focuses either on the effects on the receiving country as a whole, or on outcomes caused by regional differences in immigrant exposure. Human capital, however, moves not only across but also within countries. Although the assimilation of foreigners advances over time, cultural traits, and production know-how are remarkably persistent across generations (Abramitzky et al., 2014; Valencia, 2019; Voth, 2021). Foreign settlers attract further immigrants through networks and, as their spatial diffusion is gradual, the incidence of the original inflow unfolds unevenly across regions and cohorts. Resulting long-run effects of mass immigration on other, initially unpopulated regions, however, are not yet well understood.

In this paper, we examine how the inter-regional spread of historical immigrants’ descendants relates to contemporary labor market outcomes in Brazil. We use linked employer–employee data from 2008 to 2019 and apply a surname-based machine learning algorithm to classify workers’ ancestries.¹ We distinguish between workers with Iberian surnames (hereafter *locals*), who constitute the predominant group in the population, and descendants of non-Iberian immigrants (*descendants*). Our central research question is how variation in the local concentration of descendants is associated with wage levels across municipalities.

The two labels “locals” vs. “descendants” are shorthand and should not be read as describing culturally homogeneous groups. “Locals” is in fact a residual category comprising the descendants of Iberian settlers, of enslaved Africans and of Indigenous populations who all carry Iberian surnames today. “Descendants” is broad in a different way, covering the non-Iberian an-

¹For the purposes of this study, we define “ancestry” as the country of origin of one’s ancestors.

cestries our classification distinguishes. What separates the two is thus not homogeneity within each of them but provenance. The backgrounds pooled under the first label were formed in Brazil over four centuries of Iberian settlement, slavery and Indigenous presence, while those pooled under the second arrived from Europe, the Levant and Japan between 1850 and 1960 and settled in relatively self-contained colonies. It is this difference in provenance, and not any ranking of the two groups, that make differences and complementarity between them plausible.

Brazil provides a particularly suitable setting to study the long-run consequences of migration, as migrants played a central role in two major episodes that shaped its economic geography. Between 1850 and 1960, roughly five million immigrants arrived to work in the emerging manufacturing industries of the Southeast and in agricultural production in the South. About 45% of these immigrants were of non-Iberian origin—including Italian, German, Syrian-Lebanese, and Japanese migrants—altering the demographic composition of a country historically shaped by Indigenous populations and Iberian settlement (IBGE, 2007). These inflows may also have influenced the local stock and spatial distribution of human capital in and around immigrant settlement areas (Carvalho Filho and Monasterio, 2012; Ehrl and Monasterio, 2024; Rocha et al., 2017).

A second migration episode began around 1960, when millions of internal migrants moved from the coastal and southern regions toward the interior in the “March to the West” (Pellegrina and Sotelo, forthcoming). Facilitated by infrastructure expansion (Bird and Straub, 2020), land grants and colonization programs (Jepson, 2006a; Hosono and Hongo, 2012), and technological innovations that enabled large-scale agriculture in previously unproductive tropical areas (Bragança et al., 2015), this movement populated regions that had not been directly shaped by first-generation immigrants.

These two migration waves define the empirical setting of this study. We distinguish between the states that hosted the historical immigrant settlements and the remaining states, where non-Iberian immigration was negligible and where most descendants present today arrived through internal migration after 1960. At the same time, international immigration to Brazil declined sharply and has since remained negligible, leaving the contemporary foreign-born share close to zero. Workers with non-Iberian surnames in our data therefore overwhelmingly represent Brazilian-born descendants rather than recent immigrants. This setting enables us to examine how the inter-regional diffusion of inherited cultural human capital is associated with current

labor market outcomes by comparing these two distinct regional contexts.

Focusing on wages as the outcome variable permits an analysis at the individual level with more precise identification. Aggregate outcomes like average wages can mix composition and spillovers effects. For example, if descendants bring higher human capital to a labor market and are paid higher wages, they will raise the average wage of that market by increasing the share of high relative to low wages, not necessarily by raising wages for all workers in that market. On the other hand, if descendants bring human capital that is complementary to the existing human capital of non-descendants, there might be positive spillovers to all or only a specific group of workers. Using individual-level outcomes like wages, therefore, allows us to separate composition and spillover effects. Additionally, in an individual-level analysis we can explore several heterogeneities in our results including (but not limited to) different effects the concentration of descendants can have on the non-descendants and on the descendants themselves. Finally, we leverage the richness of our panel data and include fixed effects at the individual level to account for effects from spatial sorting, productivity related characteristics, discrimination, and the direct impact of individual ancestry itself. We combine the fixed effects wage regressions with an instrumental variable approach as in Combes et al. (2008); De la Roca and Puga (2017); Ehrl and Monasterio (2024) to address remaining endogeneity issues from reverse causality and measurement error. The instrumental variable is the distance to the location of official colonial settlements. We argue that this variable captures the presence of immigrant descendants in interior regions shaped by the expansion of the agricultural frontier, while being plausibly unrelated to general wage levels. Greater distance from the historical settlement areas is associated with a lower likelihood of observing descendants, but we do not expect distance itself to affect wages other than through differences in immigrant presence.

The data show that the concentration of (non-Iberian) descendants in the local workforce is positively and significantly related to the average wage level of municipalities in the states with little historical immigration, whereas there is no comparable relation in the South and Southeast, where immigrants concentrated historically. Still, there are remarkable differences between ancestry groups. Wages of workers in the predominant group are higher where descendants are more concentrated, whereas workers with a non-Iberian ancestry show no such difference. In the two-step specification with individual fixed effects, one additional percentage point in the concentration

of descendants corresponds to a 0.94% higher wage for non-descendants. Instrumenting the descendant share with the distance to the historical colonies preserves this sign and yields a larger coefficient, although we treat that magnitude with caution for the reasons discussed in Section 4.

We use a simple model with a standard production function and labor demand and supply, adapted from studies that investigate imperfect substitution between groups of workers (Borjas et al., 2008; Ottaviano and Peri, 2012), to organize the discussion of the mechanisms behind our results. The framework is deliberately used *ex post*, as an interpretive device rather than as a source of predictions that the empirical analysis then tests. Its role is to separate the part of the observed association that follows from the composition of the workforce alone—factor complementarity for locals and diminishing marginal returns for descendants—from the part that requires a productivity spillover. Read through this lens, the combination of no wage difference within the descendant group and a sizable positive one for locals points to positive spillovers from the presence of the scarce non-Iberian ancestry group.

This study contributes to three strands of literature. First, we add to the literature on the long-term consequences of the Age of Mass Migration (Hatton and Williamson, 1998). Several studies document positive link between European immigration and development in the US and Latin America (Sánchez-Alonso, 2007; Droller, 2018; Valencia, 2019; Sequeira et al., 2020). Those studies are usually constrained to the original sites of colonization. One rare example is Von Berlepsch and Rodríguez-Pose (2021) who show that destinations of second-generation US migrants currently have a higher income per capita. By focusing on how the descendants of historical immigrants spread their human capital to other regions in a receiving country and affect its labor markets, our study broadens the current knowledge on the long-term impacts of historical events.

Second, because the internal migration flows of historical immigrants' descendants in Brazil are closely tied to the expansion of modern mechanized agriculture in the country, we contribute to the literature on the causes and consequences of the expansion of the agricultural frontier in Brazil (Bragança, 2018; Bustos et al., 2016, 2020; Pellegrina, 2022). Anecdotal accounts often mention how instrumental the descendants were in shaping the modern profile of agriculture and the economies along the frontier (Wagner and Bernardi, 1995; Alves, 2005; Rezende, 2002; Alves, 2016). However, no study directly identifies the descendants of historical migrants and attempts to analyze the impacts of their human capital in these regions as we do. More broadly, the human capital of

migrants can be seen as another factor shaping the settlement of the frontier, alongside geography (Nunn and Puga, 2012; Bazzi et al., 2016) and the internalized frontier experience (Bazzi et al., 2020).

Third, we add to the literature that investigates imperfect substitution between domestic and immigrant labor (Borjas and Katz, 2007; Ottaviano and Peri, 2012), extending the analysis to the *descendants* of these workers. In doing so, we also connect to the literature on immigrants' assimilation (Abramitzky et al., 2014; Pérez, 2021). Non-Iberian descendants being imperfect substitutes for the predominant population group in the labor market implies that intangible assets are transmitted over generations in line with evidence surveyed in Bisin and Verdier (2011) and Voth (2021). Emergence of productive spillovers through combination of cultural backgrounds aligns with knowledge sharing, complementarity of approaches, problem solving as documented by research on the value of immigrant diversity (Ozgen, 2021).

The rest of the paper is organized as follows. Section 2 provides background information on immigration, ancestries, and surnames in Brazil. This section discusses our samples of municipalities, and presents a brief account of the expansion of the agricultural frontier in Brazil. Section 3 presents the data, explains the surname-based classification of ancestries, and shows some descriptive statistics. Section 4 presents the empirical strategy, discusses identification concerns and proposed mitigation approaches. Section 5 shows the results and robustness checks. Section 6 contains our theoretical framework used to rationalize the empirical patterns. Section 7 concludes the paper.

2 Background information

2.1 Immigration, surnames, and ancestries in Brazil

The colonial ties of Brazil to Portugal and its proximity to the Spanish colonies in South America resulted in a regular flow of colonizers and immigrants coming from the Iberian Peninsula. This process gave Portuguese and/or Spanish ancestries—and surnames—to most of Brazilians who declare themselves as ‘whites’.² At the same time, Brazil’s historical – and many times forceful

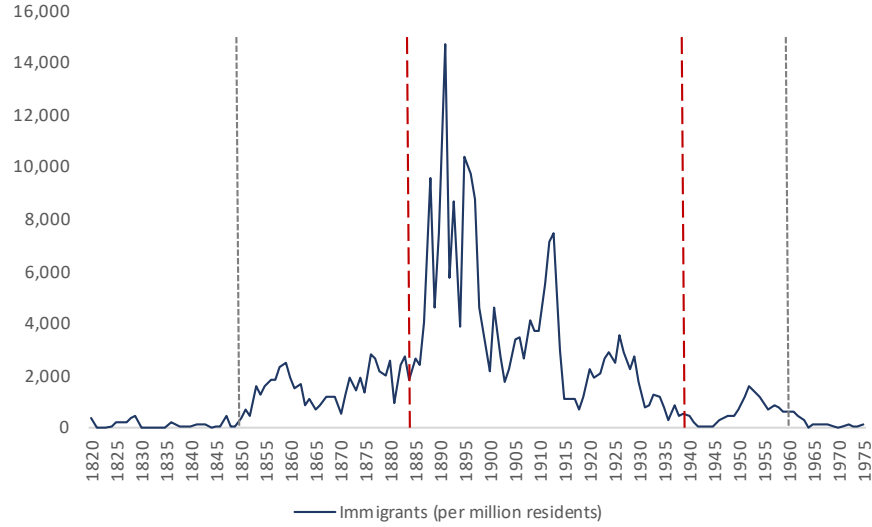
²Official census surveys and administrative data in Brazil ask people to determine their skin color/race into one of five possible categories: white, mixed (*pardo*), far-eastern (*amarelo*) and native-Brazilian (*indígena*). These categories, however are often far too broad and imprecise to allow for specific applications like ours.

– integration of former slaves and Amerindians into its national population, left the descendants of those groups with Iberian surnames as well. As a result, not only Brazilian whites but also virtually all Brazilian blacks, mixed, and natives have Iberian surnames today.

From the mid-nineteenth century through the early twentieth century, Brazil actively encouraged international immigration as part of a broader development strategy. Beginning in the 1850s and continuing through the 1930s, the government promoted the arrival of European and other foreign settlers, both to foster settlement and agricultural development in the interior and to replace slave labor following abolition. As a result, Brazil became one of the destinations of the Age of Mass Migration. Inflows intensified from the late nineteenth century through the late 1950s (with a temporary decline during World War II) and then fell sharply after 1960 (Figure 1). During the peak period of 1884–1939, 45% of the recorded immigrants came from non-Iberian countries and 43% from Portugal and Spain, while the origin of the remaining 12% is not separately recorded, according to Census compilations reported in IBGE (2007). The largest non-Iberian groups originated from Italy, Germany, and Japan, with smaller inflows from Syria, Lebanon, Turkey, Poland, Russia, and other countries. Immigration from Portugal and Spain, by contrast, occurred throughout Brazil’s history and was more geographically widespread. Table A1 provides the full breakdown by country of origin. With the sharp decline in new inflows after 1960 and the natural aging of earlier cohorts, the foreign-born share of the Brazilian population has since become negligible, amounting to only 0.23% in the 2010 census.

Brazil’s historical background, combined with this intense – but later interrupted – experience of international immigration in its post-slavery period, generated a unique landscape of surnames and ancestries in the country. Because the fraction of foreign-born in the country today is close to zero and because most Brazilians have Iberian surnames, a person bearing a non-Iberian surname in Brazil has a high probability of having descended from immigrants who arrived in the country between 1850 and 1960. Therefore, the surname-based classification of ancestries employed in our analysis constitutes an imperfect but useful proxy for the presence of descendants of historical immigrants in the current population through the origin of the surname. This approach has been applied in previous studies in Brazil and other contexts, which document its validity in capturing ancestry-related variation (Güell et al., 2015; Ehrl and Monasterio, 2024; Posch et al., forthcoming).

Figure 1: International immigration in Brazil, 1820–1975



Note: Immigration and resident numbers from IBGE (2007). The gray lines delimit the 1850–1960 period, whereas the red lines delimit the period for which we have more information on immigrants’ country of origin (1884–1939), see Table A1.

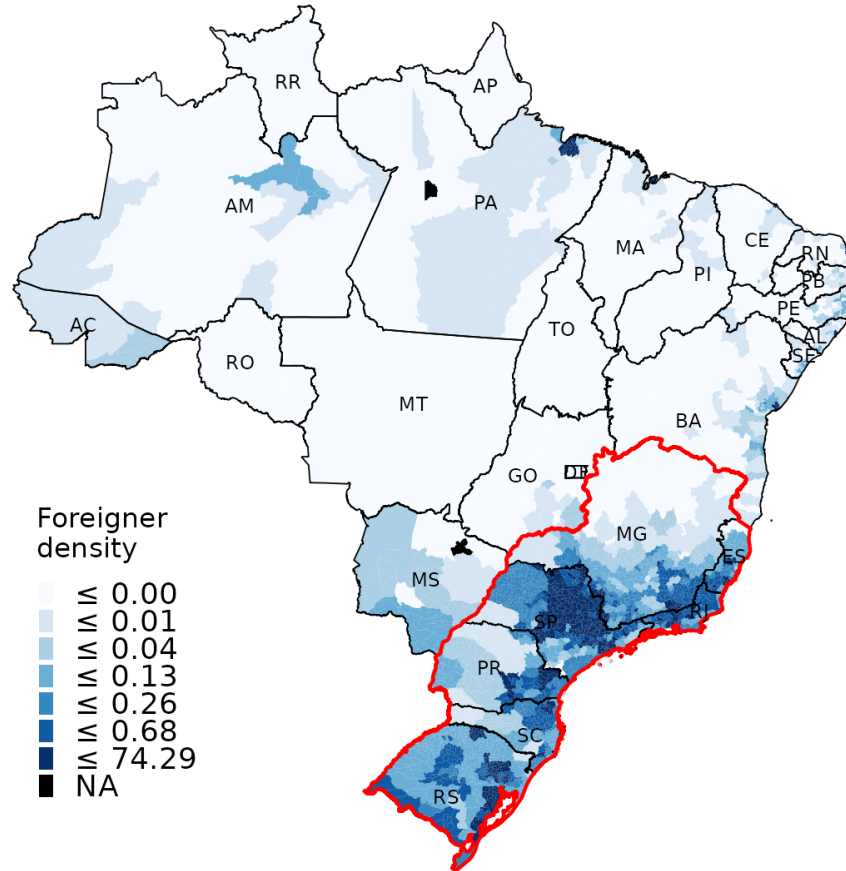
2.2 Study regions – high and low historical immigration samples

Several states in Brazil have well documented clusters (injection points) of historical non-Iberian immigration. Particularly the state-sponsored settlements (known as *colônias*) attracted a large number of non-Iberian immigrants to the South and Southeast regions of Brazil. There are historical records of non-Iberian settlements in the states of Minas Gerais (Monteiro, 1973), Espírito Santo (Franceschetto, 2014), São Paulo (Rocha et al., 2017), as well as for the three states in the South region: Paraná (Nishikawa, 2015), Santa Catarina (Piazza, 1983), and Rio Grande do Sul (Carvalho Filho and Monasterio, 2012).³ These immigrant clusters used to be unpopulated before 1850 and thus developed slowly (Roche, 1969; Bublitz, 2008). Once installed, the immigrants spread only gradually to nearby regions, creating a pattern of spatial persistence (Ehrl and Monasterio, 2021).

Figure 2 shows the density of foreign immigrants in 1920 – approximately the apex of first-generation immigrants in Brazil. The map illustrates the first of two background observations that motivate our sample split. Both are documented in the sources cited above and we restate them here because our empirical design rests on them.

³There are occasional historical records of non-Iberian settlements in states of the other three regions in Brazil (e.g. some Japanese settlements in the states of Amazonas and Pará in the North region). However, the records of settlements in these regions are rare compared to other regions, and the settlements are smaller.

Figure 2: Foreigner density in 1920 and sample definition



Notes: The map shows the number of foreign immigrants in the year 1920 per square kilometer for each of the Minimum Comparable Areas (AMC). AMCs are an aggregation of municipalities such that the defined area remains stable over the 1920–2020 period, according to Ehrl (2017). Black lines denote state boundaries. The *low historical immigration sample* encompasses all municipalities/AMC in the states of Rondônia (RO), Acre (AC), Amazonas (AM), Roraima (RR), Pará (PA), Amapá (AP), and Tocantins (TO) in the North region; the states of Maranhão (MA), Piauí (PI), Ceará (CE), Rio Grande do Norte (RN), Paraíba (PB), Pernambuco (PE), Alagoas (AL), Sergipe (SE), and Bahia (BA) in the Northeast region; and the states of Mato Grosso do Sul (MS), Mato Grosso (MT), and Goiás (GO) in the Center-West region. The *high historical settlements sample* is formed by the municipalities in the states of Rio Grande do Sul (RS), Santa Catarina, and Paraná (PR) in the South region, and the states of São Paulo (SP), Espírito Santo (ES), Rio de Janeiro (RJ), and Minas Gerais (MG) in the Southeast region—all of which have a significant and well documented presence of injection points of immigration, are at or very close to the sea, and thus present a high foreigner density. Legend categories correspond to deciles of foreigner density; the lowest are aggregated due to high frequency of near-zero values.

Stylized Fact 1. *By 1920, foreigners were heavily concentrated in the South and Southeast of Brazil, in and around the original injection points of non-Iberian immigration.*

Internal migrants from the South and Southeast increasingly settled in the Center-West during the 1960s and 1970s and later in parts of the North and Northeast following the adaptation

of soybean varieties to tropical conditions. This process marked the large-scale expansion of the agricultural frontier and transformed previously sparsely populated areas into economically active regions. Frontier development was driven primarily by modern agricultural production, but also integrated local markets, stimulated service provision and agro-industrial activities, accelerated the diffusion of new technologies, and reshaped land use and regional economic structures (Bragança et al., 2015; Bustos et al., 2016, 2020). Settlement was further facilitated by private colonization companies, cooperatives, land reform initiatives, and federal development programs (Jepson, 2006a,b).

Motivated by these distinct settlement trajectories, we distinguish between two groups of municipalities (Figure 2). The first group, which we refer to as the *high historical settlements sample*, comprises the municipalities of the seven states with documented nineteenth- and early twentieth-century concentrations of non-Iberian immigrants: Rio Grande do Sul, Santa Catarina and Paraná in the South, and São Paulo, Espírito Santo, Rio de Janeiro and Minas Gerais in the Southeast. The second group, the *low historical immigration sample*, comprises the municipalities of the remaining states, in the North, Northeast and Center-West regions.

We define the two samples along state borders, and thus as contiguous territories, for two reasons. First, the historical records of official settlements are organized by state, so the presence or absence of documented injection points is a state-level attribute in our sources (Section 2.1). Second, a contiguous partition keeps the two groups internally comparable in terms of the regional labor markets, transport networks and state-level institutions that their municipalities share. The label of the second sample refers to the historical inflow of non-Iberian immigrants, which was negligible throughout these states, and not to the age of settlement: parts of the Northeast are among the longest-settled areas of Brazil, whereas the demographic and economic expansion of the interior of the North and Center-West took place after 1960 with the advance of the agricultural frontier, including the broader “March to the West” movement (Pellegrina and Sotelo, forthcoming). What the municipalities of this sample have in common, and what our analysis exploits, is that whatever descendants they host today arrived through internal migration rather than through direct settlement by first-generation immigrants.

3 Data

3.1 Data sources

The main data source used in this study is the *Relação Anual de Informações Sociais* (RAIS), a report of all labor contracts that employers in Brazil are required to file every December in order to comply with labor regulations. This makes RAIS a high-quality annual census of all formally employed workers in Brazil.

The RAIS data include demographic characteristics of employees (race, gender, age, education), their remuneration, hours worked per week, occupation, and a unique identification number. The data also include characteristics of the employers like industry, and the municipality in which the establishment is located. The full name of workers, used in our surname-based classification of ancestries, is a key distinctive variable in the RAIS data which is available in the years from 2008 to 2019.

Since the RAIS registers all formal employment relations, it is common to make a few adjustments to restrict noise and capture jobs where the wage is determined by demand and supply in the market (Gerard et al., 2021). Therefore, we exclude all foreign-born, all public servants, and the military. We keep workers between 16 and 70 years old who have a valid identifier, work more than 10 hours per week and record a positive remuneration. After adjusting remunerations for inflation and calculating hourly wages, we drop those workers with hourly wages above the 99.9th percentile. Due to the usual adoption of the husband’s surname after marriage, as explained in the following subsection, we focus on male workers to have a more direct link between surname and the person’s cultural heritage. Finally, we exclude municipalities with less than 30 workers and those with a zero share of descendants.

We complement this data with multiple sources for municipality-level information. Socioeconomic characteristics come from Ipeadata. Rainfall variables are constructed using data from the Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS). Temperature variables use data from the Global Meteorological Forcing Dataset for land surface modeling. Average elevation and the Terrain Ruggedness Index are calculated using data from the Global Multi-resolution Terrain Elevation Data 2010 (GMTED2010) from the US Geological Survey. The geolocation of the historical non-Iberian settlements in São Paulo and Rio Grande do Sul come

from Rocha et al. (2017) and Carvalho Filho and Monasterio (2012), respectively. The distance to these non-Iberian settlements uses municipalities' economic centers obtained using Google Maps. Variables on soil types use data from Embrapa Solos, while the dummies and shares for biomes use information from the MapBiomas project.

3.2 Surname-based classification of ancestries and descendant share

This subsection describes how we classify ancestries based on workers' surnames and briefly discuss the social and legal norms surrounding surnames in Brazil.

Children in Brazil usually receive two surnames: the mother's second surname, followed by the father's second surname.⁴ Because only the second surname of each parent is passed on, and because the father's surname comes last, effectively, only the father's surname survives over generations. As for name changes after marriage, Brazilian civil law required a married woman to adopt her husband's second surname until 1977. After that, adoption became optional. In most cases, when adopted, the husband's surname becomes the second.

There are some challenges to implementing a surname classification that uses two or more surnames. In our study, we consider only the last surname of each person. We do not expect, however, a more refined classifications to improve the approximation of our measure to the true concentration of individuals of non-Iberian ancestry in the municipalities of our study samples for two reasons. First, the group of non-Iberian ancestries is large enough to accommodate many cases of mixed ancestries (e.g., German-Italian). Second, we expect neither a consistent pattern in the order of Iberian and non-Iberian surnames when a person has both, nor a correlation between this order and the concentration of descendants in a given municipality. Therefore, the measurement error arising from assigning Iberian or non-Iberian ancestry to workers with mixed surnames is probably classical in our setting (any measurement error arising from the choice of using only the second surname in our classification will bias our results towards zero).

Simply put, the surname-based classification of ancestries used in this study is a refined version of the algorithm described in Monasterio (2017), where more datasets are used to increase the number of distinct names and ancestries. Therefore, 16 different European-Asian, non-Iberian

⁴Some people in Brazil have three or more surnames. Nevertheless, here we use the term "second" to refer to the surname that comes last in a person's full name to avoid confusion with the term "last name", commonly used in English to denote one's surname.

ancestries can be identified from the following four steps: (1) collect the second surname of all workers in the RAIS; (2) match these surnames to historical immigration records where surnames are accompanied by countries of origin; (3) link each unique surname to a country of origin or region of origin (e.g., Eastern Europe); and (4) apply a machine-learning language classification method proposed by Cavnar and Trenkle (1994) to the remaining unmatched surnames. More than 96% of the distinct surnames were identified through fuzzy matching to the immigration records, and only the remainder through the method of Cavnar and Trenkle (1994).

The classification yields, for each municipality m and year t in our sample, the total number of workers of each ancestry. We group all ancestries into two categories: Iberians, i.e. locals (IBR) and non-Iberians, i.e. descendants (NIB). Our variable of interest in the empirical analyses is the concentration of descendants given as

$$C_{mt}^{Desc} = \frac{N_{mt}^{NIB}}{N_{mt}^{NIB} + N_{mt}^{IBR}} \quad (1)$$

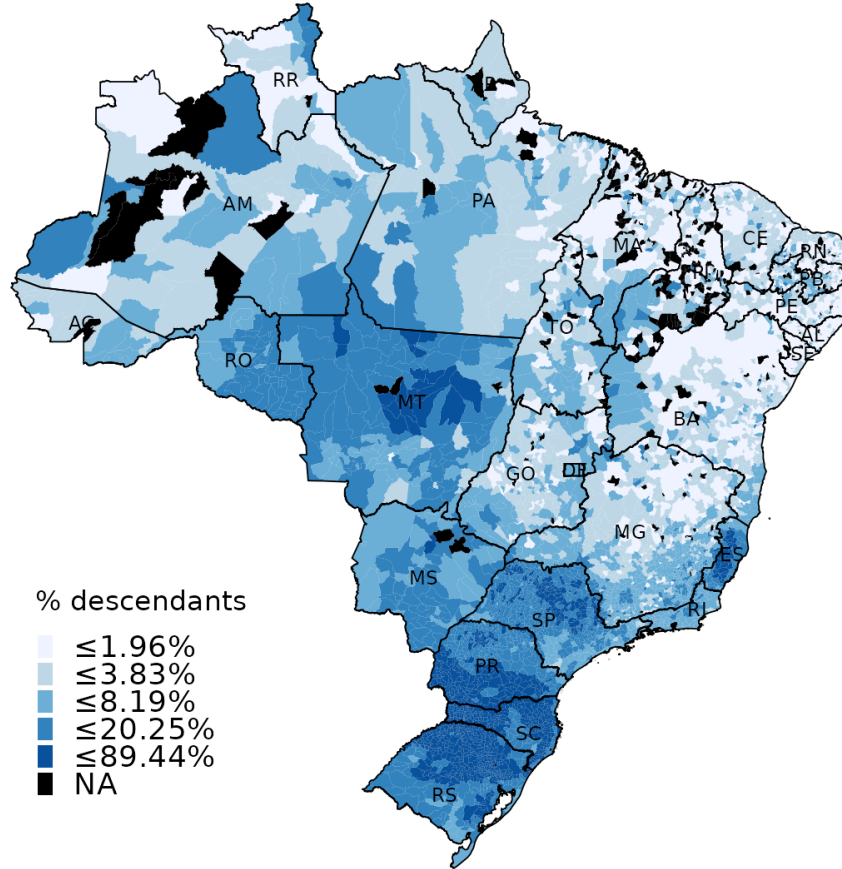
Figure 3 shows the spatial distribution of the concentration of non-Iberian descendants across Brazilian municipalities in 2008. In approximately half of the Southern municipalities the share of non-Iberian descendants is between 20% and 89%. Moreover, there is a high concentrations of descendants in São Paulo (SP) and Espírito Santo (ES), all places that are known for having received a significant number of historical non-Iberian immigrants. We also notice a higher concentration in the Center-West region, particularly in Mato Grosso, a state that symbolizes the expansion and modernization of the agricultural frontier in Brazil. The Northwestern states tend to register the lowest concentration of descendants, roughly half of them not having more than 2%.

Complementary to the spatial distribution, Figure A1 in the Appendix shows the density distribution of the concentration of descendants across the municipalities in our two samples. Both distributions are skewed to the left, but beyond the share of 5%, the distribution in the high historical settlements sample is clearly the dominant one.

Stylized Fact 2. *Currently, descendants of foreign immigrants are heavily concentrated in the South and Southeast of Brazil, mirroring the historical pattern of immigrant settlement.*

Together with stylized fact 1, this second background observation establishes the spatial persistence on which our sample split relies. The stylized facts that follow, in Sections 3.3 and 5,

Figure 3: Concentration of descendants in Brazilian municipalities, 2008 (%)



Notes: The concentration of descendants is given by the percentage of workers with a non-Iberian surname in the formal workforce in each municipality in 2008. The scale in the map uses quintiles to highlight variation in the concentration across space. Black lines denote state boundaries. The denomination of states and their regions are: Rondônia (RO), Acre (AC), Amazonas (AM), Roraima (RR), Pará (PA), Amapá (AP), and Tocantins (TO) in the North region; the states of Maranhão (MA), Piauí (PI), Ceará (CE), Rio Grande do Norte (RN), Paraíba (PB), Pernambuco (PE), Alagoas (AL), Sergipe (SE), and Bahia (BA) in the Northeast region; Mato Grosso do Sul (MS), Mato Grosso (MT), and Goiás (GO) in the Center-West region; Rio Grande do Sul (RS), Santa Catarina, and Paraná (PR) in the South region, and the states of São Paulo (SP), Espírito Santo (ES), Rio de Janeiro (RJ), and Minas Gerais (MG) in the Southeast region.

summarize our own estimates.

3.3 Descriptive statistics

This section by presents descriptive statistics to test for differences between our samples, and for differences between descendants and locals within each sample. Table 1 shows the mean for several variables of interest from the RAIS data and additional information from the Census in 2010. Due

to the large sample sizes, all differences are statistically significant. Therefore, we present the normalized difference next to the differences in parentheses (Imbens and Wooldridge, 2009). A common rule of thumb is to consider a normalized difference greater than a quarter as meaningful.

Comparing the two samples indicates that the number of workers per municipality as well as the average population size is about thrice as large in the high historical settlements sample. The degree of labor formalization is also considerably higher there (about 71% against 57%), which means that informality and self-employment are more prevalent in the low historical immigration sample. In line, other indicators for economic development and socioeconomic welfare such as years of schooling, life expectancy, and average wage are also considerably higher in the high historical settlements sample (Appendix Table A3). The share of industrial activity is lower in the low historical immigration sample, while agricultural share is higher. Finally, we observe that the average concentration of descendants is much higher in the high historical settlements sample, formally confirming our stylized fact 2.

Table 1: Summary Statistics, Individual Level, 2010

	high historical settlements sample				low historical immigration sample			
	Locals	Descendants	Diff. (L – D)		Locals	Descendants	Diff. (L – D)	
Panel A: Information from RAIS data								
Individuals per municipality (1,000)	477	332	145	[0.14]	118	107	12	[0.06]
Share of non-Iberian descendants	13.1%	21.9%	-8.7%	[0.52]	3.7%	6.1%	-2.4%	[0.58]
Log of hourly wage	2.44	2.66	-0.23	[0.21]	2.15	2.45	-0.30	[0.33]
Hours worked	42.46	42.94	-0.48	[0.08]	43.05	43.34	-0.29	[0.06]
Age (years)	34.61	35.19	-0.58	[0.04]	33.81	34.54	-0.73	[0.05]
Education level: HS or higher	0.52	0.63	-0.12	[0.17]	0.50	0.61	-0.12	[0.16]
White collar occupation	0.15	0.25	-0.10	[0.18]	0.12	0.23	-0.11	[0.24]
Industry = Agriculture	0.05	0.04	0.01	[0.03]	0.08	0.11	-0.03	[0.08]
Industry = MFG/Construction/Utility	0.38	0.40	-0.02	[0.03]	0.37	0.31	0.06	[0.08]
Panel B: Information from Census data								
Total population (1,000)	2,575	1,777	798	[0.15]	862	773	89	[0.07]
Expected years of schooling	10.00	10.18	-0.19	[0.21]	9.51	9.57	-0.07	[0.07]
Life expectancy at birth (years)	76.00	76.17	-0.17	[0.11]	73.97	74.39	-0.42	[0.16]
Degree of labor formalization	70.6%	71.6%	-1.0%	[0.09]	56.7%	58.3%	-1.7%	[0.10]
Individuals (Millions)	12.4M	2.1M	14.5M		5.8M	0.2M	6M	

Notes: The high historical settlements sample includes municipalities in the South and Southeast regions. The low historical immigration sample includes municipalities in the Center-West, North, and Northeast. Both samples exclude municipalities with fewer than 30 individual observations in our selected RAIS data or no descendants. Indicators in the top panel use individual-level data from RAIS in 2010; those in the bottom panel use municipality-level data from the Brazilian population census in 2010. The share of non-Iberian descendants is the % of workers with a non-Iberian surname in the municipality where the worker is employed, averaged over workers. Due to the large number of observations in all samples, all differences are statistically significant at 1%. Therefore, we show the absolute value of the normalized differences, which are unaffected by sample size, in brackets.

Regarding the two types of workers, we observe that the wage of descendants is significantly higher than the wage of locals in both samples and that this difference is more pronounced in the low historical immigration sample.⁵ Another fact, aligned with conjectures about the ‘March to the West’ is that descendants are less likely to work in the agricultural sector in the high historical settlements sample, whereas they are more likely to work in that sector in the low historical immigration sample. Table 1 further indicates that descendants and locals differ in demographic characteristics. In particular, descendants are more educated, have a higher life expectancy, and concentrated in white collar occupations. The differences hold with similar normalized magnitudes and the same directions in both samples. While inherently difficult to measure, persistent elements of cultural heritage or work ethic may constitute an additional productive asset that distinguishes between the groups of locals and descendants (Spolaore and Wacziarg, 2013; Voth, 2021).

Stylized Fact 3. *Non-Iberian descendants and locals exhibit systematical differences in observable labor-market characteristics and outcomes.*

4 Empirical framework

The overall aim of this paper is to describe how historical immigration relates to contemporary local labor-market outcomes. Specifically, we ask whether the local share of immigrants’ descendants is systematically associated with workers’ wages. Throughout the paper we use the language of associations. The successive adjustments described below—individual fixed effects, municipality-level controls, and instrumentation—narrow the set of explanations that can account for the estimated coefficients, but none of them delivers a causal design.

Consider the following regression equation at the municipality level as an initial motivation for our empirical analysis.

$$y_{m(s)t} = \beta C_{mt} + Z_{mt}\Gamma + \psi_s + \phi_t + \varepsilon_{m(s)t} \quad (2)$$

$y_{m(s)t}$ is the average log wage in municipality m located in state s in year t , C_{mt} is the contemporary concentration of descendants in the municipality, Z_{mt} is a vector of municipality-level character-

⁵We do not find differences in hours worked, which indicates that using hourly wages or total labor income as the outcome of interest should produce similar results.

istics, ψ_s and ϕ_t are state and time fixed effects, respectively, and $\varepsilon_{m(s)t}$ is the error term. The estimated coefficient β shows the approximate percentage increase in wages associated with one additional percentage point in the share of descendants in the municipality. We run this equation separately for the historical settlements and the low historical immigration sample. Moreover, we investigate whether the effects differ between immigrants' *descendants* and the remainder of the Iberian workers (*locals*).

Several identification threats in equation (2) preclude a causal interpretation of the estimated coefficients. (1) The share of non-Iberian descendants is naturally imprecise because some people may have a cultural legacy from a non-Iberian country which is not (anymore) reflected in their surname. Likewise, recent Spanish and Portuguese immigrants cannot be distinguished from traditionally Brazilian residents because the linguistic origin is the same. This measurement error tends to bias β towards zero. (2) Reverse causality may be present when either locals or descendants systematically sort into high-wage municipalities. In fact, Ehrl and Monasterio (2024) find that non-Iberian immigrants' descendants are less mobile today, such that we can expect (further) downward bias in the estimated β from equation (2). (3) Attributes of municipalities like first-nature advantages and those shaped by historical immigrants affect the wage level and will bias our estimates if not properly controlled for. Political institutions and the educational system, for example, have been positively affected by the immigrants in the 19th and 20th century (Seyferth, 1994; Naritomi et al., 2012).

To deal with these threats to identification, our empirical strategy makes the following adjustments. We adopt a two-stage estimation procedure as in Combes et al. (2008), Bakens et al. (2013), De la Roca and Puga (2017), Ehrl and Monasterio (2021), among others. Exploiting the panel nature of our individual data, we run a first-stage wage regression on the exogenous time-varying worker characteristics X_{it} (age and age²) and fixed effects for individuals (μ_i), municipalities (κ_m), and years.

$$y_{i(m)t} = X_{it}\delta' + \phi_t + \kappa_m + \mu_i + \varepsilon_{i(m)t} \quad (3)$$

The individual fixed effects account for possible bias from sorting on unobservable attributes and also nets out the influence of the worker's *own* cultural background. Consequently, the estimated municipality fixed effect $\hat{\kappa}_k$ captures remaining wage differences between the local labor markets.

The second-stage regression is then given by

$$\hat{\kappa}_m = \beta C_m + Z_m \Gamma + \xi_m \quad (4)$$

That is, through the first-stage estimation we account for any dynamic effects from the composition of workers with the heterogeneous cultural backgrounds and abilities. The vector Z_m has several controls that correlate with first- and second nature advantages like agricultural productivity, distance to the sea, and population density. Precisely, Z_m includes measures for total yearly rainfall and the annual average temperature in the municipality (average and standard deviation in the 1981–2010 period). It also includes the total population in the municipality and its area (both in logs), a state capital indicator, and indicators for soil types.⁶ Remaining wage differences due to the share of descendants C_m thus emerge as a form of external effect from the production function.

To address remaining endogeneity issues from omitted variable bias, measurement error, and reverse causality in equation (4), we apply an instrumental variable (IV) approach. Following Ehrl and Monasterio (2024), our instrument is the average distance from each municipality to former official colonies located in the states of Rio Grande do Sul and São Paulo (the settlements discussed in Section 2.1).⁷ The instrument should be relevant because of the spatial persistence of immigrants documented in Figures 2 and 3, and because staying close to the historic injection points reduces migration costs and eases network formation. We thus expect the colony distance IV to be negatively correlated with the concentration of descendants. The instrument should also satisfy the exclusion restriction because the *official* colonies in the two states were chosen by government officials based on military and strategic settlement considerations. Their location is thus unrelated to first-nature advantages and was not endogenously determined by immigrants' choices, see Ehrl and Monasterio (2024) for further details and motivation about the IV.

⁶One may argue that estimating equations with municipality fixed effects would strengthen the control for time-constant amenities that immigrants installed. However, on the one hand, the main variable of interest (C_m) has very little temporal variation, making this strategy infeasible. A variance decomposition of the descendant share in the municipality level for the years 2008–2019 reveals that 98% of the variable's variation is across units and only 2% across time. On the other hand, the instrumental variable strategy is already intended to isolate the pure effect of descendant concentration on wages. Since the instrument also has no variation over time, we can only provide these estimations in a cross-section of municipalities.

⁷Although other states in the high historical settlements sample also had settlements, we consider only the settlements in São Paulo and Rio Grande do Sul when calculating colony distance measures. These are the states for which there is evidence of long-term impacts in the economic literature, and for which there is information on the current municipality that corresponds to the location of the original settlements in these states (Carvalho Filho and Monasterio, 2012; Ehrl and Monasterio, 2024).

Two qualifications are in order. First, because the official colonies lie in the South and Southeast, colony distance varies along a broad north–south gradient. That gradient is precisely the one along which the settlement of the interior proceeded, which is what makes the instrument relevant, but it also means that we cannot rule out that distance is correlated with other slowly varying determinants of local wage levels.⁸ Second, the exclusion restriction of a just-identified model cannot be tested directly, and we therefore treat the instrumented estimates as complements to, rather than replacements for, the fixed-effects estimates.

We prefer IV models with a single instrument because exactly identified IV estimators exhibit limited bias, particularly when the first-stage coefficient’s sign accords with theory (Angrist and Kolesár, 2024). To obtain at least a partial check on exogeneity, we add a second instrument, the Terrain Ruggedness Index. The rationale behind it is that the suitability of potential destinations for modern agriculture (low terrain ruggedness) worked as a pulling factor for descendants contemplating a move to the agricultural frontier in Brazil since the 1960s. As we explained in Section 2.2, regions with a high availability of flat farmland disproportionately attracted non-Iberian descendants with the expertise and means to engage in modern agriculture, coming from areas (the high historical settlements sample) where agricultural production is characterized by high productivity and capital usage (Bustos et al., 2020). We thus expect average terrain ruggedness to negatively correlate with the concentration of descendants. Note that ruggedness enters here as a shifter of settlement and migration decisions, not as a direct determinant of local productivity: conditional on colony distance and on the geographic and climatic controls in Z_m , we do not expect it to shift wage levels through other channels. The Hansen J test it makes available asks whether the two instruments deliver mutually consistent estimates, and thus corroborates the exclusion restriction rather than establishing it. Figure A2 in the Appendix gives a visual representation of the two instruments.

⁸For the same reason we do not include state fixed effects in the second-stage regressions. Colony distance is measured from the economic centers of municipalities to a fixed set of settlements in two southern states, so it varies far more across states than within them: regressing the instrument on state dummies yields an R^2 of 0.985. Conditioning on state would discard almost all of the variation the instrument provides.

5 Results

5.1 Municipality-level Correlations

In Table 2, we show the results for municipality-level regressions of wages on the concentration of descendants according to equation (2). In all specifications, we account for regional differences between states through fixed effects and for observable first- and second-nature advantages of municipalities. Columns (1) to (3) refer to the high historical settlements sample, whereas columns (4) to (6) are based on the low historical immigration sample of municipalities farther away from the locations of historic immigrants. For each sample, we estimate panel regressions using the years 2008-2019, as well as regressions using only the first or last year available.

Table 2: Municipality-Level Regressions of Wages on the Concentration of Descendants

	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent variable: log wage					
	high historical settlements sample			low historical immigration sample		
Years	2008–19	2008	2019	2008–19	2008	2019
Non-Iberian Descendants share	0.031 (0.092)	0.011 (0.101)	0.031 (0.086)	1.251*** (0.314)	1.169*** (0.337)	1.479*** (0.282)
Mean of descendants	0.15	0.15	0.14	0.04	0.04	0.04
Adj. R ²	0.83	0.61	0.64	0.90	0.83	0.82
Observations	34,126	2,835	2,849	30,386	2,467	2,566
Municipalities	2,855	2,835	2,849	2,689	2,467	2,566
State fixed effects	Y	Y	Y	Y	Y	Y
Time fixed effects	Y			Y		

Notes: The dependent variable is the log of the average formal sector wage in the municipality. The non-Iberian descendant share (%) represents the share of workers with a non-Iberian surname among the total number of formal employments in the municipality. All regressions are weighted by the number of workers in the municipality and include the following municipality-level controls: historical average (1981–2010) and standard deviation of total yearly rainfall and average temperature, total population (log), municipality area (log), and five dummies for soil types (dummies = 1 if 5% or more of municipality area is covered by soil type). Clustered standard errors at the municipality-level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The results reveal a consistent positive association between the concentration of descendants and wages in the low historical immigration sample. In contrast, the null of no significant correlation cannot be rejected in the high historical settlements sample. The mean descendant share in each specification, reported below the coefficients in the table, confirms that municipalities in the

South and South-East (0.15) have more than 3 times more non-Iberian descendants compared to the low historical immigration sample, with tiny or no variation over time.⁹ Column (6) indicates that one additional percentage point in the concentration of descendants is associated with 1.48% higher average wage in the low historical immigration sample. We summarize these findings in the following stylized fact.

Stylized Fact 4. *There is a positive association between the concentration of descendants and average wages in municipalities, however, only for municipalities in the low historical immigration sample where descendants are much more scarce.*

5.2 Two-step regressions

Table 3 shows the results from the second step according to equation (4). The first major difference with respect to the previous results in Table 2 is that regressions are performed separately for descendants and locals. Second, the first-step estimations account for worker fixed effects and thus absorb the influence of workers' own ancestry on wages as well as spatial sorting on individuals' characteristics, see equation (3). Since the previous estimates were consistent across time (and continue to be so, see Table A2 for the year 2008), we focus on the year 2019 for the rest of the paper.

The results for the high historical settlements sample tell a story of no effects for the descendants and relatively small but positive effects for the other population group, locals. In the low historical immigration sample, the relation between the concentrations of descendants and local workers' wages is also positive but about 3.5 times larger. Yet, for the own population group, a higher share of descendants seems to depress wages in the low historical immigration sample.

Regarding the differences between the two specifications in Tables 2 and 3, we observe that the adjustment through the first-step regression leads to a lower correlation between the share of descendants and wages. The marginal effect of an additional percentage point in the concentration of descendants is now associated with a 0.94% higher wage for locals in the low historical immigration sample. This coefficient is still highly significant, below the 1% level.¹⁰ The

⁹The comparison across years indicates that, due to the lack of temporal variation, panel regressions will not be more informative than cross-section specifications.

¹⁰Our results are not sensitive to using other type of standard errors, including bootstrap or Conley procedures. Because of adding the additional instrumentation stage in the following estimations, we prefer to report the simple,

Table 3: Two-step Wage Regressions – Descendants vs. Locals, 2019

	(1)	(2)	(3)	(4)
Dependent Variable:	Estimated municipality fixed effect			
Sample:	historical settlements		low historical immigration	
Years:	2019	2019	2019	2019
Workers:	Descendants	Locals	Descendants	Locals
Non-Iberian descendants share	0.023 (0.029)	0.266*** (0.032)	-0.288*** (0.093)	0.939*** (0.165)
Mean of Descendants	0.14	0.14	0.04	0.04
Adj. R ²	0.60	0.48	0.45	0.58
Observations	2,841	2,845	2,518	2,562

Notes: The table reports coefficients from the two-step wage regressions according to equations (3) and (4). Non-Iberian descendant share (%) represents the share of workers with a non-Iberian surname among the total number of formal employments in the municipality. All regressions are weighted by the number of workers, either locals or descendants, according to the fifth row, in the municipality and include the following municipality-level controls: historical average (1981–2010) and standard deviation of total yearly rainfall and average temperature, total population (log), municipality area (log), and five dummies for soil types (dummies = 1 if 5% or more of municipality area is covered by soil type). The controls in the first-stage worker-level wage regressions include age, age squared, as well as municipality, worker, and year fixed effects. Clustered standard errors at the municipality-level in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

reduced coefficient in the two-step procedure indicates that without accounting for sorting based on intrinsic worker attributes, the estimates are biased upwards. That is, workers with abilities that are particularly valuable in the labor market sort into high-wage regions. The direction and relative magnitude of the implied bias is comparable to the approach in De la Roca and Puga (2017).

5.3 Two-step regressions with instrumentation

In this subsection we complement the two-step estimates with an instrumental variable (IV) strategy that speaks to the remaining endogeneity concerns. Table 4 reports IV regressions for 2019 in the low historical immigration sample. Columns (1) and (4) instrument the share of descendants with the average distance to the historical colonies in Rio Grande do Sul and São Paulo, columns (2) and (5) with terrain ruggedness alone, and columns (3) and (6) with both. The upper part of the table reports the second-stage coefficient, the lower part the first-stage coefficients and the associated 2SLS statistics.

transparent clustered standard errors.

Table 4: Two-step Wage Regressions with instrumentation – Descendants vs. Locals

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable:	Estimated municipality fixed effect					
Sample	low historical immigration					
Years	2019	2019	2019	2019	2019	2019
Workers	Descendants			Locals		
Instrument	Colony dist.	Ruggedness	Both	Colony dist.	Ruggedness	Both
Non-Iberian	0.366	0.574	0.391	3.032***	6.370	3.321***
Descendants share	(0.830)	(1.082)	(0.645)	(1.042)	(4.563)	(0.994)
Observations	2,518	2,518	2,518	2,562	2,562	2,562
IV 1.-Stage Statistics						
Colony Distance	-0.029*** (0.004)		-0.030*** (0.004)	-0.025*** (0.004)		-0.024*** (0.003)
Terrain Ruggedness		-0.003 (0.002)	-0.003*** (0.001)		-0.001 (0.001)	-0.002*** (0.001)
1. F-Stat	46.527	2.289	31.682	46.164	1.116	27.581
R ² -partial	0.224	0.025	0.263	0.291	0.007	0.289
Weak IV AR	0.207	0.252	0.799	6.578**	3.476*	12.147***
Weak IV SW	0.345	0.367	2.178	10.660***	4.542**	33.687***
Hansen J (overid)			0.014			1.019
Hansen J (p-value)			0.907			0.313

Notes: The table reports coefficients from the two-step wage regressions with instrumentation according to equations (3) and (4). The dependent variable thus is the estimated municipality fixed effect obtained from the first-stage worker-level wage regressions that also include age, age squared, worker and year fixed effects as controls. The non-Iberian descendant share (%) represents the share of workers with a non-Iberian surname among the total number of formal employments in the municipality. Columns (1) and (4) instrument the descendant share with the average distance to the historical non-Iberian settlements in the states of São Paulo and Rio Grande do Sul; columns (2) and (5) instrument it with the average Terrain Ruggedness Index alone, which is weak on its own and reported only for completeness; columns (3) and (6) use both instruments and are the specifications for which the overidentification test is defined. This average distance to settlements (expressed in 1,000km) is an average of the distances from the economic center of a given municipality to the economic center of all municipalities with a non-Iberian historical settlement in the states of Rio Grande do Sul and São Paulo as recorded by (Carvalho Filho and Monasterio, 2012) and (Rocha et al., 2017), respectively. The second IV is the average Terrain Ruggedness Index (TRI), which was calculated using the methodology proposed by (Riley et al., 1999), and topographical data from the Global Multi-resolution Terrain Elevation Data 2010 (GMTED2010) from the US Geological Survey (resolution: 15 arc-seconds). The index calculates the difference in elevation between a grid cell and its surroundings. Higher values of the index correspond to more rugged terrain. We average the index values for all grid cells in a municipality to obtain its average TRI. The low historical immigration sample includes all municipalities in the Center-West, North, and Northeast with at least one descendant and 30 or more individual observations in our selected RAIS data. The sample excludes females. All regressions are weighted by the number of workers, either locals or descendants according to the fifth row, in the municipality. The lower part of the table reports the following first-stage IV statistics: IV coefficient and standard error, excluded IV F-statistic and partial R², weak IV Anderson–Rubin (AR) Wald and Stock–Wright LM test statistic, Hansen–Sargan overidentification test for exogeneity of IVs and its p-value. Clustered standard errors at the municipality-level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Column (4) shows that the positive association between the concentration of workers with non-Iberian ancestry and the wages of locals remains significant with the instrumentation. The

coefficient is more than three times the one obtained without instrumentation, the direction expected if measurement error and sorting of workers with Iberian ancestry bias the uninstrumented estimate downward. Taken at face value, an additional percentage point in the concentration of descendants corresponds to a 3% higher wage for locals. We read this as an upper bound and place the weight of our interpretation on the sign, which is the same in Tables 2, 3 and 4. Column (1) suggests that descendants' wages are not systematically related to a higher concentration of their own group.

The first-stage statistics confirm that colony distance is a relevant instrument. Its coefficient is negative, as expected, highly significant, and similar in both worker samples. Terrain ruggedness is correctly signed once colony distance is controlled for, but with first-stage F statistics of 2.29 and 1.12 it is weak on its own, so we do not interpret columns (2) and (5); distance to the previous settlements evidently accounts for the bulk of the variation in descendant concentration along the agricultural frontier. The Anderson–Rubin and Stock–Wright statistics, which remain informative when instruments are weak, reject the joint null of no effect and valid instruments for local workers but not for descendants.

Columns (3) and (6) use both instruments and thus allow a Hansen J test. Its outcome does not reject the joint exogeneity of the two IVs, which corroborates the exclusion restriction without establishing it, and the estimated coefficients remain close to those in columns (1) and (4).

Stylized Fact 5. *A higher concentration of non-Iberian descendants is positively associated with the wages of the Iberian population only, and not with the wages of the descendants themselves.*

5.4 Robustness checks

Focus on ancestry in the formal labor market. One concern with the use of RAIS data to classify surnames is that it covers only a share of the population in a given municipality-year: the formally employed. In a country like Brazil, where informality is high, a potential concern for the present analysis is that the share of non-Iberian descendants in the total labor market is quite different the measure in the RAIS data. To address this shortcoming, we resort to a couple of other data sources that also inform respondents' names in Brazil: the *Cadastro Único*, the unified registry of beneficiaries of the Brazilian cash transfer program (*Bolsa Família*), and the *Base Sócios*, a record of business owners maintained by the national tax authority. For at least one year (2010), we

can compare the concentration of descendants obtained using only RAIS or using these three data sources combined, which cover a larger portion of the population in the municipalities we study. The correlation between the two measures across all Brazilian municipalities in 2010 is positive and very high ($r = 0.96$). In figure A3 in the appendix we plot the concentration of descendants using only RAIS data on the same concentration using the extended data sources. The results show a concentration of observations around the 45° line, reassuring us that the concentration of descendants we calculate using only surnames of formally employed workers is a good representation of the true concentration of descendants in the population of the municipalities in our samples.

Sample composition. Table A4 in the Appendix presents the robustness checks regarding sample composition and institutional differences between municipalities, for the two-step wage regressions with and without instrumentation. On the one hand, we check if our results are not influenced by municipalities with the highest or lowest number of individual observations. In columns (1) and (4), we restrict the sample to within the 5th and 95th percentile of the distribution of observations per municipality weighted by the number of workers (i.e. locals). On the other hand, we exclude municipalities (in the low historical immigration sample) with an elevated concentration of foreign immigrants in the year 1920. We exclude all municipalities with a foreigner density above 0.04, see Figure 2. This test serves to confirm that the municipalities with a high concentration of current descendants were not historically different from other regions. Finding divergent results from the baseline sample would also be misaligned with our interpretation that it was primarily the internal migration beginning in the 1960s that shaped the labor markets in the low historical immigration municipalities.

Controls for institutions and human capital. Following previous papers such as Naritomi et al. (2012), we test whether local institutional quality affects the results. The potential threat to identification is that non-Iberian descendants either chose municipalities with a high institutional quality—similar to their origin—or even shaped the public landscape over time. Similarly, another main difference that may drive our results could stem from differences in the educational system. Therefore, the estimations in Appendix Table A4, columns (3) and (6), include proxy variables for governmental quality, access to justice, and a proxy for educational performance: test results by 9th graders in a nationwide exam.

Results across the six specifications in Table A4 are consistent with our main results for local workers. The association between the concentration of descendants and wages remains positive and significant at the 1% level in every column, with and without instrumentation. The estimated coefficient is somewhat smaller than in the baseline once we account for institutional quality and educational performance, which is what one would expect if part of the baseline association operates through the local institutional and educational environment rather than around it.

6 Theoretical framework and discussion

This section presents a simple formal framework to rationalize the stylized facts documented in the empirical analysis. We first outline the model and then discuss the conditions under which the observed patterns are consistent with its predictions. We place this section after the results because it is used *ex post*. The framework is not a structural model that we estimate, and it is not a set of predictions that the previous sections test. Its purpose is to make explicit which part of an observed association between the descendant share and wages follows from the composition of the workforce alone, and which part cannot be accounted for without a productivity spillover.

Our framework adapts the model of Borjas et al. (2008) and Ottaviano and Peri (2012), who analyze wage effects arising from changes in the relative concentration of immigrant and native workers. This framework may be suited to our setting because it allows for imperfect substitutability between labor types and for productivity spillovers across groups. In our adaptation, we treat descendants of non-Iberian immigrants and Brazilians without non-Iberian ancestry as distinct types of labor input.

Suppose each municipality is a competitive economy that produces a single output good, Y , using capital, K , and labor provided by non-descendants, N_0 , and descendants, N_1 . The three inputs are imperfect substitutes in the production technology of firms. For simplicity, we assume this technology has constant returns to scale and can be represented by a Cobb-Douglas production function. Imperfect substitutability is thus an assumption of the framework rather than one of its results, and the Cobb-Douglas form fixes the elasticity of substitution between the two labor types at unity. As we note below, the signs of the wage responses that we derive are preserved under any finite elasticity of substitution; only their magnitudes depend on the functional form.

$$Y = (\theta_0 N_0)^{\alpha_0} (\theta_1 N_1)^{\alpha_1} K^{1-\alpha_1-\alpha_0}, \quad \alpha_0 > 0, \alpha_1 > 0, \alpha_0 + \alpha_1 < 1. \quad (5)$$

The terms θ_0 and θ_1 are the productivity shifters of each type of labor and capture productivity spillovers in the model. We assume shifters are a function of the worker's ancestry-specific human capital, ϕ , and the spillover $f(s)$. Specifically, the model allows for human capital spillovers by letting workers' productivity depend on the share (s) of type 1 workers in each municipality. As in our empirical approach, we transform equation (5) in logarithms:

$$\log(\theta_j) = \phi_j + f(s), \quad \text{for } j = 0, 1, \quad s = \frac{N_1}{N}, \quad N = N_1 + N_0. \quad (6)$$

We assume that wages are equal to the marginal product of each type of labor and that the productivity spillovers are external to individual firms in the municipality but internal to the municipality as a whole. That is, firms take the θ 's as given and choose the amounts of labor N_0 and N_1 that maximize profits ignoring their effect on the respective productivity shifters θ_0 and θ_1 . At the same time, spillover effects are constrained to each municipality.

We relax two assumptions in the original model. First, we let both types of workers—descendants and locals here—have different productivities, $\theta_1 \neq \theta_0$, but we do not assume that the productivity of the first type is necessarily higher. Second, we do not assume monotonic spillovers. Instead, we let $f(s)$ take any format. In our setting, increasing the concentration of one worker type can have non-linear effects on the wages of both types.¹¹

The derivatives of log wages with respect to the share of type 1 workers, s , show how a change in the concentration of descendants affects the wages of both worker types:

$$\frac{d \log(w_0)}{ds} = \left[\frac{1 - \alpha_0}{1 - s} + \frac{\alpha_1}{s} \right] + (\alpha_1 + \alpha_0) f'(s) \quad (7a)$$

and

$$\frac{d \log(w_1)}{ds} = - \left[\frac{\alpha_0}{1 - s} + \frac{1 - \alpha_1}{s} \right] + (\alpha_1 + \alpha_0) f'(s). \quad (7b)$$

The wage changes in both equations are composed of two different effects. The first term in square brackets is a *factor complementarity* effect in (7a) and a diminishing marginal returns effect in

¹¹A higher s , thus, is not necessarily better: f could be maximized at some $s^* \in (0, 1)$.

(7b). The second term is driven by the derivative of $f(s)$, the *spillover* effect. Equation (7b) implies that if the relative size of one type in the production function increases, its respective marginal product and the associated wage must decrease. This follows from the imperfect substitutability of the two labor types and would hold for any finite elasticity of substitution between them. If the observed total association is instead positive or indistinguishable from zero, a positive spillover must be offsetting the first term. This is the one inference we draw from the framework, and it does not require us to commit to a particular value of the elasticity or to the magnitude of any of the terms. The model also implies that wages of locals increase more than those of descendants when the latter share rises: $d \log(w_0)/ds > d \log(w_1)/ds$ should thus be satisfied in any case.¹²

The stylized facts 1, 2, and 3 about the spatial concentration and distinction in labor-market characteristics between locals and descendants of foreign immigrants align with the model's assumption of treating the two groups as different types of labor input. The different magnitudes in the average wage of *all workers* in the low historical immigration and high historical settlements sample may be related to different production functions and to the differences in the level of the non-Iberian descendant share, see Table 2 and the stylized fact 4.

Once we split the sample by ancestry group in Tables 3 and 4, a positive association between the concentration of descendants and the wage of *locals* in a municipality is consistent with both complementarity and spillover effects (stylized fact 5). Because the diminishing returns effect is negative for the *descendants*, an insignificant overall effect suggests that positive spillovers compensate the former effect. The considerably high overall effect on the other population group, the workers with Iberian ancestry, also suggests that positive spillovers are at work.

Our findings are at odds with other possible explanations for the observed wage externality. Since the benefits do not accrue to all groups in the population, potential benefits from non-Iberian descendants such as a higher capital stock or public goods provision do not seem to apply in the present case.

¹²From equations (7a) and (7b) we have:

$$\frac{d \log(w_0)}{ds} - \frac{d \log(w_1)}{ds} = \frac{1}{s(1-s)} > 0.$$

7 Conclusion

Our study uses a surname-based classification of workers' ancestries to trace the imprint of two historical events on current local labor markets in Brazil. First, mass immigration from 1850 to 1960 increased the size and cultural diversity of the labor force in Brazil, but immigrants concentrated in the South and Southeast. Second, internal migration to the largely unpopulated interior of Brazil starting in the 1960s spread the presence of historical immigrants' descendants in these regions.

Our empirical analysis uses matched employer–employee panel data from 2008 to 2019 and exploits rich individual and municipality fixed effects to absorb persistent worker heterogeneity and time-invariant local characteristics. To further reduce concerns related to measurement error in the ancestry classification and potential simultaneity between local composition and wages, we complement the baseline specifications with an instrumental variable strategy. While these approaches substantially narrow the scope for confounding influences, the results should be interpreted as stylized facts of robust associations rather than as definitive causal effects.

We find that higher concentrations of descendants of historical non-Iberian immigrants in the states of the North, Northeast and Center-West—the low historical immigration sample—are systematically associated with higher wage levels. The same association appears, with a considerably smaller coefficient, in the states that hosted the historical settlements. In both cases the wage difference is confined to workers with Iberian surnames, that is, to the predominant group in the local population.

Our observations are in accordance with a simple framework with standard labor supply and demand in which the two groups of workers are imperfect substitutes and generate positive productivity spillovers. Because the diminishing-returns term works against the descendants' own wages, an association of zero within that group already requires an offsetting positive spillover; the sizable positive association for locals points in the same direction. Because we observe descendants in municipalities far from the historic injection points, our results cannot be explained by fixed factors linked to the sites of historical immigration such as land redistribution, natural endowments, or a head start in infrastructure. Therefore, we interpret our results as consistent with mobile factors having been an important channel through which the long-run correlates of historical non-Iberian immigration in Brazil spread from the injection points to other regions of the country. This

interpretation is consistent with inter-generational transmission of task-relevant cultural traits and implies that ancestry can be economically salient, especially when heterogeneous backgrounds are combined to exploit complementarities.

Our evidence speaks to the level of entire local labor markets. The fixed effects absorb the composition of the workforce and the attributes of individual workers, so what remains is a difference in the wage level that a market offers rather than a difference in who works there. This is the level at which our contribution lies, and it does not depend on which micro foundation generates it. What we document is an externality operating within the local labor market. Identifying its source is a promising direction for future work. An analysis at the establishment-level could show whether the returns arise inside firms that employ both groups or between firms in the same market, and would open the role of employers' own ancestry to investigation.

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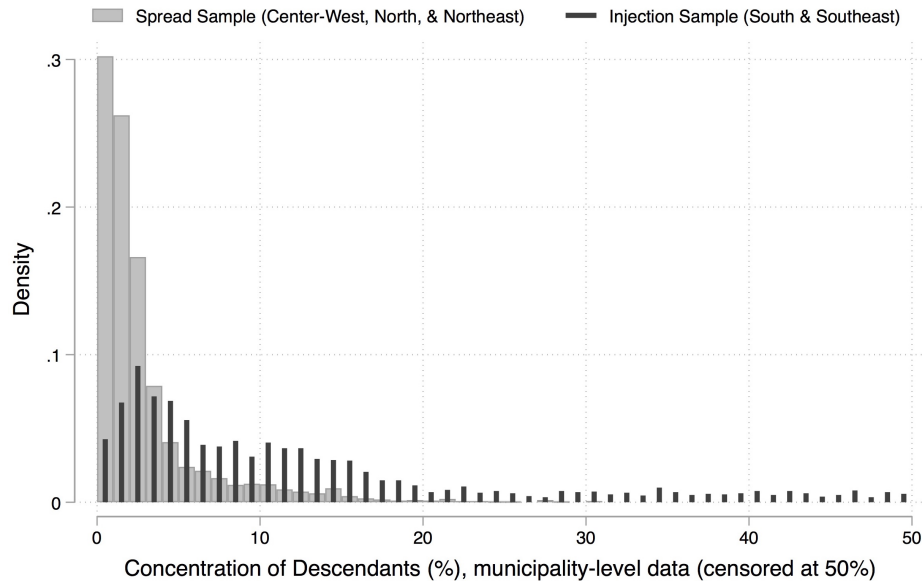
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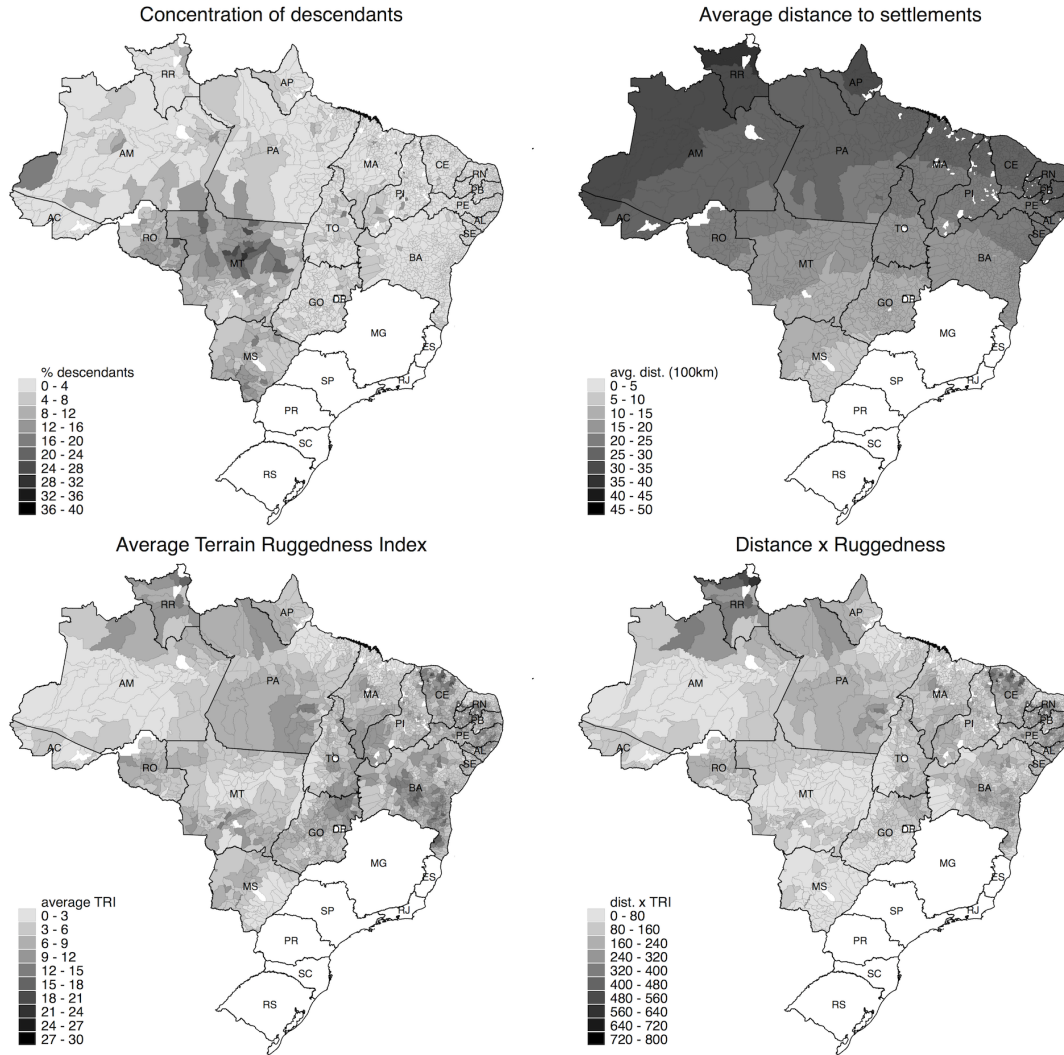
Appendix – For Online Publication

Figure A1: Distributions of the concentration of descendants in the municipalities, 2010



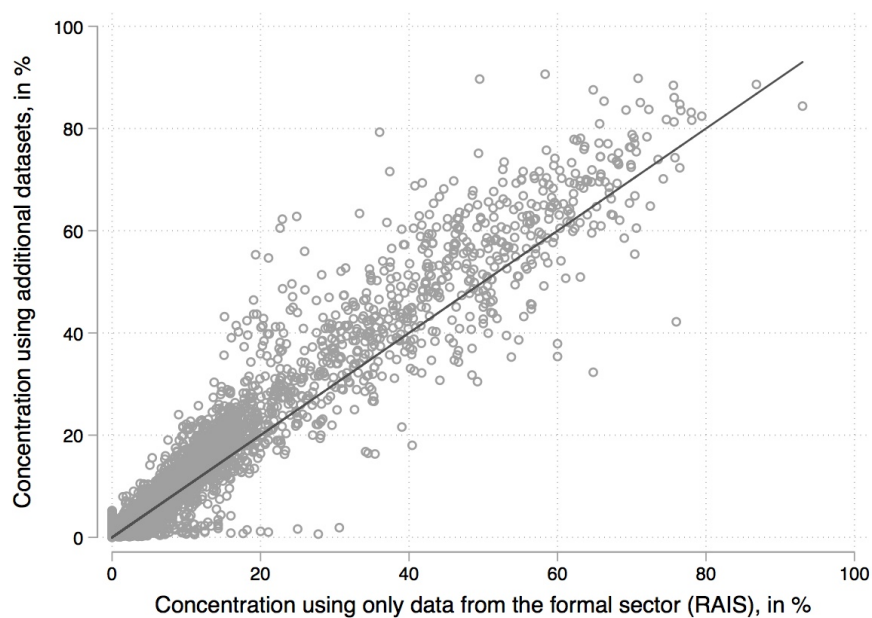
Notes: The concentration of descendants is given by the percentage of workers with a non-Iberian surname in the formal workforce in each municipality in 2010. The figure shows the distribution up to 50% only to improve visualization, since few municipalities in the high historical settlements sample and no municipality in the low historical immigration sample go beyond that level. The high historical settlements sample considers municipalities in all states of the regions South and Southeast, whereas the low historical immigration sample considers municipalities in the states of the regions Center-West, North, and Northeast. For this figure only, and unlike in the estimation samples, both groups exclude state capitals and municipalities with fewer than five individual observations in the RAIS data in 2010 or with a missing value for the concentration of descendants, which is why the averages reported below differ slightly from those in Tables 1 and 2. For the high historical settlements sample, the average concentration is 16.45%, the median is 9.89%, and the standard deviation is 17.68%. For the low historical immigration sample, the average concentration is 2.88%, the median is 1.75%, and the standard deviation is 3.82%.

Figure A2: The concentration of descendants and the two instruments (distance to injection points and Terrain Ruggedness Index)



Notes: The first map (top left) shows the concentration of descendants in the study region in 2010, the measure we are instrumenting for. The second map (top right) shows the average of the distances from each municipality to the injection points of historical non-Iberian immigration in the states of São Paulo and Rio Grande do Sul. The third map (bottom left) shows the terrain ruggedness index for the municipalities in our sample. The fourth map (bottom right) shows the product of the two normalized measures and is included only to illustrate where they overlap; the two instruments enter the estimations linearly and separately, and no interaction term is used. The intervals for the scale in each graph are approximately equal to one standard deviation of each variable.

Figure A3: Concentration of descendants in the municipalities calculated using different datasets, 2010 (all Brazilian municipalities)



Note: The concentration of descendants in the horizontal axis uses only data from RAIS, while the one in the vertical axis includes also information from the *Cadastro Único* and the *Base Sócios* datasets. The graphs also include a 45° line.

Table A1: Immigration to Brazil by country of origin, 1884–1939

Country of origin	Immigrants (1,000)	Share	Ancestry group	Group share
Italian	1,412	34%	Non-Iberian	45%
Japanese	186	4%		
German	171	4%		
Syrian-Lebanese	99	2%		
Portuguese	1,204	29%	Iberian	43%
Spanish	582	14%		
Others	505	12%	Undefined	12%
Total	4,159	100%		100%

Notes: Data from IBGE (2007). Iberian refers to the sum of immigrants from Portugal and Spain.

Table A2: Two-step Wage Regressions – Descendants vs. Locals, 2008

	(1)	(2)	(3)	(4)
Dependent Variable:	Estimated municipality fixed effect			
Sample:	historical settlements		low historical immigration	
Years:	2008	2008	2008	2008
Workers:	Descendants	Locals	Descendants	Locals
Descendants share	0.038 (0.028)	0.258*** (0.032)	-0.198** (0.084)	0.847*** (0.143)
Mean of Descendants	0.15	0.15	0.04	0.04
Adj. R ²	0.59	0.46	0.52	0.61
Observations	2,823	2,827	2,376	2,461

Notes: The table reports coefficients from the two-step wage regressions according to equations (3) and (4). Descendants (%) represents the share of workers with a non-Iberian surname among the total number of formal employments in the municipality. All regressions are weighted by the number of workers, either locals or descendants, according to the fifth row, in the municipality and include the following municipality-level controls: historical average (1981–2010) and standard deviation of total yearly rainfall and average temperature, total population (log), municipality area (log), and five dummies for soil types (dummies = 1 if 5% or more of municipality area is covered by soil type). The controls in the first-stage worker-level wage regressions include age, age squared, as well as municipality, worker, and year fixed effects. Clustered standard errors at the municipality-level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A3: OLS Regressions: Indicators of Socioeconomic Development at the municipality-level, 2010 (historical settlements sample)

	(1)	(2)	(3)
<i>Panel A: Income outcomes</i>	Log of income per capita	Unemployment rate	Gini index
Concentration of descendants (%)	0.0228*** (0.0059)	-0.0067 (0.0073)	0.0090 (0.0063)
R ² (adjusted)	0.60	0.26	0.35
<i>Panel B: Education outcomes</i>	Years of schooling	HS degree or higher	Adult literacy rate
Concentration of descendants (%)	0.0075 (0.0069)	0.0207*** (0.0062)	0.0352*** (0.0056)
R ² (adjusted)	0.38	0.33	0.62
<i>Panel C: Health outcomes</i>	Life expectancy	Infant mortality	Fertility
Concentration of descendants (%)	0.0149** (0.0054)	-0.0117* (0.0051)	-0.0076 (0.0058)
R ² (adjusted)	0.54	0.55	0.50
<i>Panel D: Formal sector outcomes</i>	Log earnings (Census)	Log wage (RAIS)	Munic. wage premia
Concentration of descendants (%)	0.0154* (0.0066)	0.0624*** (0.0140)	0.0295*** (0.0073)
R ² (adjusted)	0.41	0.26	0.40
N (municipalities)	2,624		

Notes: The socio-economic indicators in panels A, B, and C were retrieved from the Atlas Brazil project (atlasbrasil.org.br/2013/en/). They reflect information from the 2010 Brazilian population census. The log earnings of the formally hired (first row in panel D) were retrieved from the 2010 census. The average log wage and the average local wage premia shown in the last two rows in panel D were calculated by the authors using information from RAIS in 2010. The measure of log earnings in the formal sector considers only hired workers with a formal labor contract. The municipality wage premium is the municipality fixed effect in a log-wage regression that includes the same extensive set of individual-level covariates used later in the main analyses. All dependent variables were standardized to facilitate the interpretation of regression coefficients. The concentration of descendants is given by the percentage of workers with a non-Iberian surname in the formal workforce in each municipality. All specifications include state fixed effects and the following municipality-level controls: historical average (1981–2010) and standard deviation of total yearly rainfall and average temperature, population density in 1950, municipality area (log), distance to the state capital (log), and dummies for biomes and soil types (dummies = 1 if 5% or more of municipality area is covered by soil type/biome). Robust standard errors in parentheses. Stars denote: * p<0.10; ** p<0.05; *** p<0.01.

Table A4: Two-step Wage Regressions with and without instrumentation – Robustness

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable:	Estimated municipality fixed effect					
Sample	low historical immigration					
Years	2019	2019	2019	2019	2019	2019
Workers	Locals					
Instrument	None			Colony Distance		
Check	5-95 pct.	low 1920 foreigners	institution controls	5-95 pct.	low 1920 foreigners	institution controls
Non-Iberian	0.694***	0.661***	0.713***	3.722***	4.892***	2.567***
Descendants share	(0.106)	(0.172)	(0.312)	(1.073)	(1.244)	(0.739)
Observations	2,312	2,428	2,451	2,312	2,428	2,451

Notes: The table reports coefficients from the two-step wage regressions with instrumentation according to equations (3) and (4). The dependent variable thus is the estimated municipality fixed effect obtained from the first-stage worker-level wage regressions that also include age, age squared, worker and year fixed effects as controls. The non-Iberian descendant share (%) represents the share of workers with a non-Iberian surname among the total number of formal employments in the municipality. The descendant share is instrumented by the average of the distance to historical non-Iberian settlements in the states of São Paulo and Rio Grande do Sul in columns (4) to (6). The low historical immigration sample includes all municipalities in the Center-West, North, and Northeast with at least one descendant and 30 or more individual observations in our selected RAIS data. The sample excludes females. All regressions are weighted by the number of workers in the municipality. The robustness checks for the estimation with and without IV are the following: columns (1) and (4) exclude municipalities outside the 5th and 95th percentile of the distribution of locals per municipality; Columns (2) and (5) exclude municipalities with a foreigner density in 1920 above 0.04 (the 50th decile); Columns (3) and (6) include proxy variables for governmental quality and access to justice from Naritomi et al. (2012), and the educational performance of 9th graders in the *Prova Brasil*, a nationwide exam conducted by INEP (the National Institute of Educational Studies and Research). Clustered standard errors at the municipality-level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$